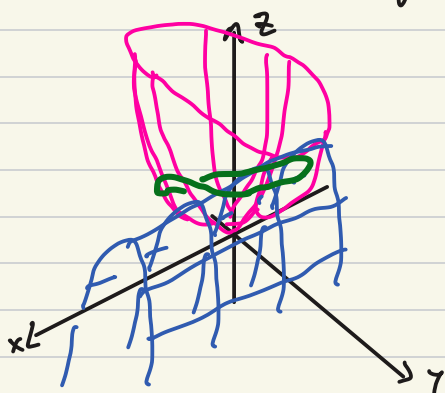


April 8, 2010

e.g. Find the volume of the region $R \subseteq \mathbb{R}^3$

bounded by $z = 4x^2 + y^2$ and $y^2 + z = 2$



↓
paraboloid
but elongated
along y

↓
 $z = 2 - y^2$
downward
parabola in the
y-z plane
and extends along
x

Let's find points of intersection of the two surfaces

$$z = 4x^2 + y^2 = 2 - y^2 \quad \leadsto \quad 4x^2 = 2 - 2y^2$$

$$\leadsto \quad 2x^2 = 1 - y^2$$

$$2x^2 + y^2 = 1$$

The intersection is an ellipse

in the x-y plane the intersection of the two surfaces
encloses the region

$$D = \{ (x, y) \in \mathbb{R}^2 : -1 \leq y \leq 1, -\sqrt{\frac{1-y^2}{2}} \leq x \leq \sqrt{\frac{1-y^2}{2}} \}$$

$$\text{then } R = \{ (x, y, z) \in \mathbb{R}^3 : -1 \leq y \leq 1, -\sqrt{\frac{1-y^2}{2}} \leq x \leq \sqrt{\frac{1-y^2}{2}}, 4x^2 + y^2 \leq z \leq 2 - y^2 \}$$

$$V = \iiint_R 1 \, dV = \int_{-1}^1 \left[\int_{-\sqrt{\frac{1-y^2}{2}}}^{\sqrt{\frac{1-y^2}{2}}} \left[\int_{4x^2+y^2}^{2-y^2} 1 \, dz \right] dx \right] dy$$

$$= \int_{-1}^1 \left[\int_{-\sqrt{\frac{1-y^2}{2}}}^{\sqrt{\frac{1-y^2}{2}}} (2 - y^2) - (4x^2 + y^2) \, dx \right] dy = \int_{-1}^1 \left[\int_{-\sqrt{\frac{1-y^2}{2}}}^{\sqrt{\frac{1-y^2}{2}}} 2 - 2y^2 - 4x^2 \, dx \right] dy$$

$$= \int_{-1}^1 \left[2x - 2y^2x - \frac{4}{3}x^3 \right]_{x=-\sqrt{\frac{1-y^2}{2}}}^{x=\sqrt{\frac{1-y^2}{2}}} dy = \int_{-1}^1 (2 - 2y^2) \left(2\sqrt{\frac{1-y^2}{2}} \right) - \frac{8}{3} \left(\sqrt{\frac{1-y^2}{2}} \right)^3 dy$$

continuous in next page

$$\begin{aligned}
 V &= \iiint_R 1 \, dV = \int_{-1}^1 \left[\int_{-\sqrt{\frac{1-y^2}{2}}}^{\sqrt{\frac{1-y^2}{2}}} \left[\int_{4x^2+y^2}^{1-y^2} 1 \, dz \right] dx \right] dy \\
 &= \int_{-1}^1 \left[\int_{-\sqrt{\frac{1-y^2}{2}}}^{\sqrt{\frac{1-y^2}{2}}} (z - y^2 - (4x^2 + y^2)) \, dx \right] dy = \int_{-1}^1 \left[\int_{-\sqrt{\frac{1-y^2}{2}}}^{\sqrt{\frac{1-y^2}{2}}} (z - 2y^2 - 4x^2) \, dx \right] dy \\
 &= \int_{-1}^1 \left[2x - 2y^2 x - \frac{4}{3} x^3 \Big|_{x=-\sqrt{\frac{1-y^2}{2}}}^{x=\sqrt{\frac{1-y^2}{2}}} \right] dy = \int_{-1}^1 (2 - 2y^2) \left(2\sqrt{\frac{1-y^2}{2}} \right) - \frac{8}{3} \left(\sqrt{\frac{1-y^2}{2}} \right)^3 dy \\
 &= \int_{-1}^1 \left(4\sqrt{\frac{1-y^2}{2}} - \frac{8}{3} \left(\sqrt{\frac{1-y^2}{2}} \right)^3 \right) dy = \int_{-1}^1 4y^2 \sqrt{\frac{1-y^2}{2}} dy \\
 &\quad \left(\begin{array}{l} \text{trig. substitution } y = \sin \theta \quad dy = \cos \theta \, d\theta \\ \text{Diagram: } y = \sin \theta \text{ on } [-1, 1] \text{ with } \theta \text{ from } -\pi/2 \text{ to } \pi/2 \end{array} \right) \\
 &\rightarrow = \int_{-\pi/2}^{\pi/2} \left(\frac{4}{\sqrt{2}} \cos \theta - \frac{8}{3\sqrt{2}} \cos^3 \theta \right) \cos \theta \, d\theta = \int_{-\pi/2}^{\pi/2} \frac{4}{\sqrt{2}} \sin^2 \theta \cos \theta \, d\theta
 \end{aligned}$$

to be finished

$$\begin{aligned}
 \text{Finished after class: } &\int_{-\pi/2}^{\pi/2} \frac{4}{\sqrt{2}} \left(\cos^2 \theta - \frac{1}{3} \cos^4 \theta - \sin^2 \cos^2 \theta \right) d\theta \\
 &= \frac{4}{\sqrt{2}} \int_{-\pi/2}^{\pi/2} \cos^2 \theta - \frac{1}{3} \cos^4 \theta - (1 - \cos^2 \theta) \cos^2 \theta \, d\theta \\
 &= \frac{4}{\sqrt{2}} \int_{-\pi/2}^{\pi/2} \cos^2 \theta - \frac{1}{3} \cos^4 \theta - \cos^2 \theta + \cos^4 \theta \, d\theta = \frac{4}{\sqrt{2}} \int_{-\pi/2}^{\pi/2} \frac{2}{3} \cos^4 \theta \, d\theta
 \end{aligned}$$

$$\begin{aligned}
 \text{Let's focus on } &\int_{-\pi/2}^{\pi/2} \cos^4 \theta \, d\theta = \int_{-\pi/2}^{\pi/2} (\cos^2 \theta)^2 \, d\theta \\
 &= \int_{-\pi/2}^{\pi/2} \left(\frac{\cos(2\theta) + 1}{2} \right)^2 \, d\theta = \int_{-\pi/2}^{\pi/2} \frac{\cos^2(2\theta)}{4} + \frac{1}{2} \cos(2\theta) + \frac{1}{4} \, d\theta \\
 &= \int_{-\pi/2}^{\pi/2} \frac{\cos(4\theta) + 1}{8} + \frac{1}{2} \cos 2\theta + \frac{1}{4} \, d\theta \\
 &= \frac{1}{32} \sin(4\theta) + \frac{1}{8} \theta + \frac{1}{4} \sin(2\theta) + \frac{1}{4} \theta \Big|_{\theta=-\pi/2}^{\theta=\pi/2} \\
 &= \frac{1}{32} (0 - 0) + \frac{3}{8} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right) + \frac{1}{4} (0 - 0) = \frac{3}{8} \pi
 \end{aligned}$$

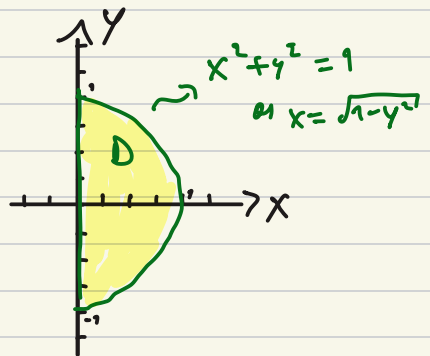
$$\text{Finally } Vol = \frac{4}{\sqrt{2}} \cdot \frac{2}{3} \cdot \frac{3}{8} \pi = \frac{\pi}{\sqrt{2}} = \sqrt{2} \pi / 2$$

More on regions of integration and order of integration

e.g. $\int_{-1}^1 \int_0^{\sqrt{1-y^2}} 3 \, dx \, dy$

$$= \iint_D 3 \, dA$$

$$= 3 \iint_D 1 \, dA$$



$$\iint_D 1 \, dA = \text{Area}(D) = \frac{1}{2} \text{Area}(\text{circle of radius } 1)$$

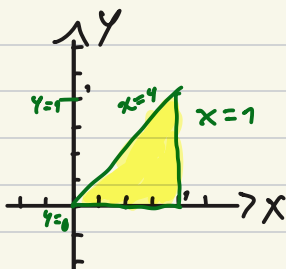
$$= \text{Area}(D) = \frac{1}{2} \pi 1^2 = \frac{\pi}{2}$$

$$\int_{-1}^1 \int_0^{\sqrt{1-y^2}} 3 \, dx \, dy = \frac{3}{2} \pi$$

e.g. $\int_0^1 \int_y^1 e^{x^2} dx dy$

x-simple $D = \{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq 1, y \leq x \leq 1\}$

y-simple $D = \{(x,y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq x\}$



$$\int_0^1 \int_0^x e^{x^2} dy dx = \int_0^1 \left[e^{x^2} y \Big|_{y=0}^{y=x} \right] dx = \int_0^1 x e^{x^2} - 0 dx$$

$$= \int_0^1 x e^{x^2} dx$$

$$u = x^2$$

$$du = 2x dx$$

$$x dx = \frac{1}{2} du$$

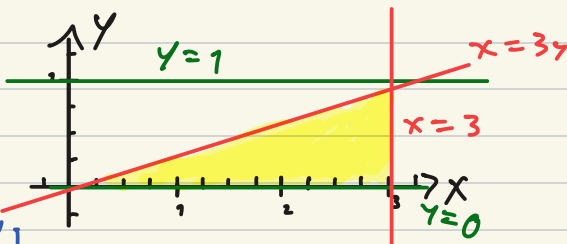
$$= \int_0^1 e^u \frac{1}{2} du$$

$$\frac{1}{2} e^u \Big|_{u=0}^{u=1} = \frac{1}{2} e - \frac{1}{2} (1) = \frac{e-1}{2}$$

e.g. $\int_0^1 \int_{3y}^3 \cos(x^2) dx dy$

x-simple $D = \{(x,y) : 0 \leq y \leq 1, 3y \leq x \leq 3\}$

y-simple $D = \{(x,y) : 0 \leq x \leq 3, 0 \leq y \leq \frac{x}{3}\}$



$$\int_0^1 \int_{3y}^3 \cos(x^2) dx dy = \int_0^3 \left[\int_0^{\frac{x}{3}} \cos(x^2) dy \right] dx$$

$$= \int_0^3 \left(\cos(x^2) y \Big|_{y=0}^{y=x/3} \right) dx = \int_0^3 \frac{x}{3} \cos(x^2) dx$$

$$u = x^2 \quad du = 2x dx$$

$$\frac{1}{2} du = x dx$$

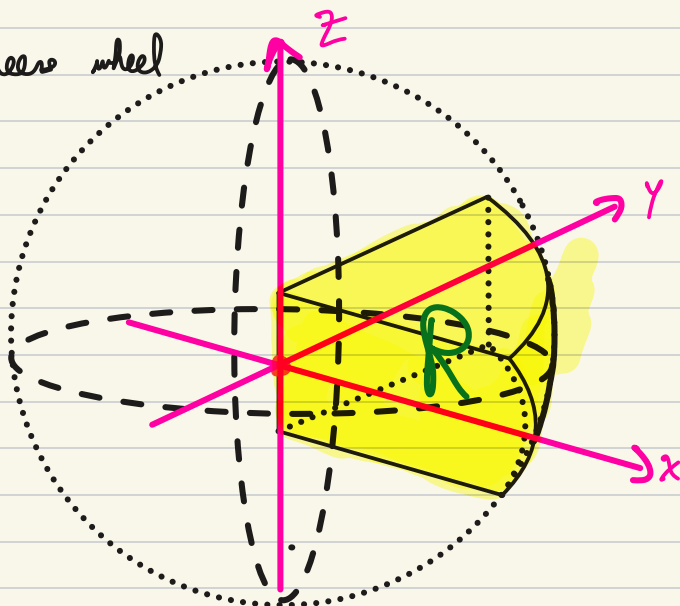
$$= \int_0^9 \frac{1}{3} \cos(u) \frac{1}{2} du$$

$$= \frac{1}{6} \sin u \Big|_{u=0}^{u=9} = \frac{\sin(9)}{6}$$

e.g. Quarter Cheddar cheese wheel

ball of rad.
= 10

height of
wheel = 8



$\iiint_R f \, dV$

set this on an iterated integral
in all the possible orders

$$R = \{(x, y, z) : -4 \leq z \leq 4, 0 \leq x \leq 10, 0 \leq y \leq \sqrt{10^2 - x^2 - z^2}\}$$

$$\iiint_R f \, dV = \int_{-4}^4 \int_0^{10} \int_0^{\sqrt{100 - x^2 - z^2}} f \, dy \, dx \, dz$$